Arabic text on LED dot-matrix displays

Mulham Fetna · ORCID 0009-0006-4432-798X

1. A problem you have already seen

Walk through any market in the region and you pass dozens of scrolling LED signs. Look closely at the Arabic ones. A great many are wrong.

Letters that should join stand apart. Words break into disconnected stumps. Some signs give up and write Arabic in Latin letters. Others display a single fixed image, produced once on a computer, which can never be changed without going back to that computer.

This is not a rare defect. It is the normal condition of the technology.

2. Why Arabic is harder than Latin here

Arabic is **cursive** and **right-to-left**, and both properties break an assumption built into these devices.

A letter's shape depends on its neighbours. Arabic letters take up to four forms — isolated, initial, medial, final. The letter ع appears as `ع` standing alone, `عـ` at the start of a word, `ـعـ` in the middle, `ـع` at the end. One character, four different shapes, chosen by context.

Certain pairs must fuse. ل followed by ا is not written as two letters; it becomes the single form `لا`. This is mandatory in correct Arabic, not a stylistic flourish.

Order and direction interact. Arabic runs right-to-left, and a line mixing Arabic with digits or Latin words needs a defined rule for what appears where.

Latin script has none of these properties. A system designed around Latin — as most display hardware is — has no mechanism for any of them.

3. Where the mismatch actually lies

Conventional sign controllers store text as **one fixed image per character code**. Character code in, stored picture out, always the same picture.

That model cannot express Arabic, because choosing the correct shape requires knowing what comes *before and after* — and a per-character lookup has no way to consult anything.

> The limitation is **architectural, not cosmetic.** It is not solved by better fonts or higher resolution, because the design has no place to ask the question Arabic requires.

This is why the problem persists in commercial products rather than having quietly been fixed.

4. Prior work — an honest assessment

It would be convenient to claim nobody has addressed this. That claim is false, and a structured survey of the field disproved it.

- **Arabic shaping on microcontrollers exists.** At least one widely used open graphics library implements contextual letter shaping and bidirectional text, and there are smaller standalone efforts as well. Credit is due, and recorded.
- **The libraries that actually drive LED dot-matrix panels do not.** They supply Arabic *glyphs* but no contextual shaping — in one prominent case the maintainer declined the feature deliberately, and user reports of disconnected letters have remained open for roughly six years.
- **One widely used plugin ships an "Arabic font"** whose own documentation states it should not be used to render ordinary Arabic text.
- **Commercially, Arabic is not a property of the sign's firmware.** Where it works at all, it is handled inside closed desktop software before anything reaches the display.

The survey is published in `related-work.md`, separating claims verified against primary sources from those that are not — including one case where automated verification returned a false negative and had to be corrected by hand.

The remaining gap is therefore narrow and specific: an open, Unicode-correct Arabic text pipeline for **low-resolution, single-colour LED dot-matrix panels**. Not "nobody solved Arabic".

5. What this project is

An open system that displays correctly shaped Arabic on inexpensive LED matrix hardware, driven from an ordinary phone, with no proprietary software anywhere in the chain.

Design goals, stated at the level of what the system guarantees:

Goal	Why it matters
Correct Arabic — proper joining and required ligatures	The point of the exercise
No app to install	Signs are operated by shopkeepers, not engineers
No internet, no router	Must work standing alone in a shop or mosque
What you preview is what appears	Operators need to trust the result before sending
Documented open protocol	Any microcontroller can implement the receiving side
Inexpensive, available hardware	Adoption depends on cost

6. Status

Working hardware, not a proposal.

Capability	Status
Driving LED matrix hardware, arbitrary panel geometry	✅ working
Self-hosted wireless operation, no router or internet	✅ working
Phone-based control interface	✅ working
Correct Arabic shaping end to end	✅ working
Static and scrolling display, with direction control	✅ working
Faithful on-screen preview	✅ working
Documented, checksummed transfer protocol	✅ working
Larger panel types	⬜ designed, not yet tested

Every capability listed as working was confirmed on physical hardware.

7. Open questions

The parts that remain genuinely unresolved, stated without inflation:

Arabic legibility at very small pixel heights. The survey found **no published work** on Arabic glyph design at the sizes these panels impose. Latin script has decades of accumulated bitmap-font craft; Arabic at comparable sizes is largely uncharted. Whether readable Arabic at this scale requires purpose-designed letterforms — rather than reduction of existing typefaces — is an open research question, and one this project is positioned to investigate.

Consistency across different devices. Ensuring identical output regardless of which device drives the sign raises questions that are understood but not fully closed.

Whether the overall approach is novel. No prior art was found for the specific architecture used here — but no exhaustive search has been performed either, so **novelty is not claimed.** Establishing this properly is recorded as outstanding work.

Quantifying the central engineering trade-off. A widely respected maintainer judged one plausible approach infeasible on resource grounds. That judgement has never been measured empirically. Turning received wisdom into evidence is outstanding work.

8. Openness and citation

Every closed alternative locks Arabic support inside desktop software that cannot be inspected, extended, or run on the device itself.

This project is released under **AGPL-3.0-or-later**, so derivatives remain open — including when operated as a network service. It is archived with a DOI so it can be cited rather than merely downloaded.

9. Summary

Arabic on LED signs is usually wrong because those devices store one fixed picture per character, while Arabic letters change shape according to their neighbours. The limitation is architectural, so it is not fixed by better fonts.

This project builds an open system that renders correctly shaped Arabic on inexpensive LED matrix hardware, controlled from an ordinary phone with no app, no router and no internet. It is working hardware with a documented open protocol, targeting a gap that a survey of prior work confirmed is real: **an open, Unicode-correct Arabic pipeline for low-resolution single-colour LED matrices.**

The most interesting question it raises is not an engineering one. It is typographic: **what does Arabic need to look like to stay readable when there are almost no pixels left to draw it with?** Nobody appears to have published an answer.